California Petroleum Systems & Thermal Basin Study

Eel River-Point Arena focus area

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1. Executive summary

The Eel River-Point Arena assignment spans a transition between the southern Cascadia forearc and the northern California transform margin. It therefore contains two petroleum settings that require separate source, reservoir and thermal interpretations. This report uses BOEM assessment outlines for the offshore comparison and the exposed Humboldt basin for onshore Eel River evidence. A planning area is an administrative reporting unit and cannot be treated as a geological basin [1]-[3].

The strongest accumulation evidence is onshore Eel River gas production from Rio Dell sandstones. Offshore, methane seeps and structures that focus fluid flow establish migration and leakage, while exploration results provide substantially weaker evidence of retained commercial accumulations. Point Arena contains oil shows and assessed sandstone and fractured Monterey-equivalent plays, but the available evidence does not establish a uniformly charged reservoir fairway [1], [2], [8].

The working model couples organic-bearing marine deposition with subsequent burial, followed by movement through permeable beds and faults into structural or stratigraphic traps. Deformation can create closure and also breach it. Candidate thermogenic source intervals include Yager rocks and Neogene marine mudstones, while shallow sedimentary organic matter can support microbial methane. The relative contributions and the age of charge remain unresolved at individual offshore prospects [5], [6].

The thermal anchor is ODP Hole 1019C. Three downhole determinations plus bottom water support a published shallow gradient of **57°C/km** and a measurement-based heat-flow estimate of **57 mW/m²**. Those results characterize the upper tens of metres below the seafloor at one location. They do not directly determine deep reservoir temperature in either basin [7].

The supported geothermal conclusion is a preliminary screening result: subsurface heat is documented, but a productive hydrothermal reservoir or commercial geothermal resource has not been demonstrated by the reviewed evidence. Deep equilibrium temperatures, hydraulic connectivity and sustainable fluid production are the decisive missing tests. The largest petroleum uncertainty is how a dated, geochemically identified source kitchen connects to a trap that has survived continuing deformation.

Report organization: main text pp. 1-11; numbered references and the AI/evidence audit follow. Observations, interpretations and proposed tests are identified throughout. The source inventory distinguishes original-document inspection from abstract-only access.

2. Study area and geological setting

The adopted focus area is the paired Eel River and Point Arena geological/assessment region shown schematically below. The Mendocino plate-boundary transition separates their tectonic contexts. The map locates the study; it does not digitize basin margins or define a legal offshore boundary [1], [11].

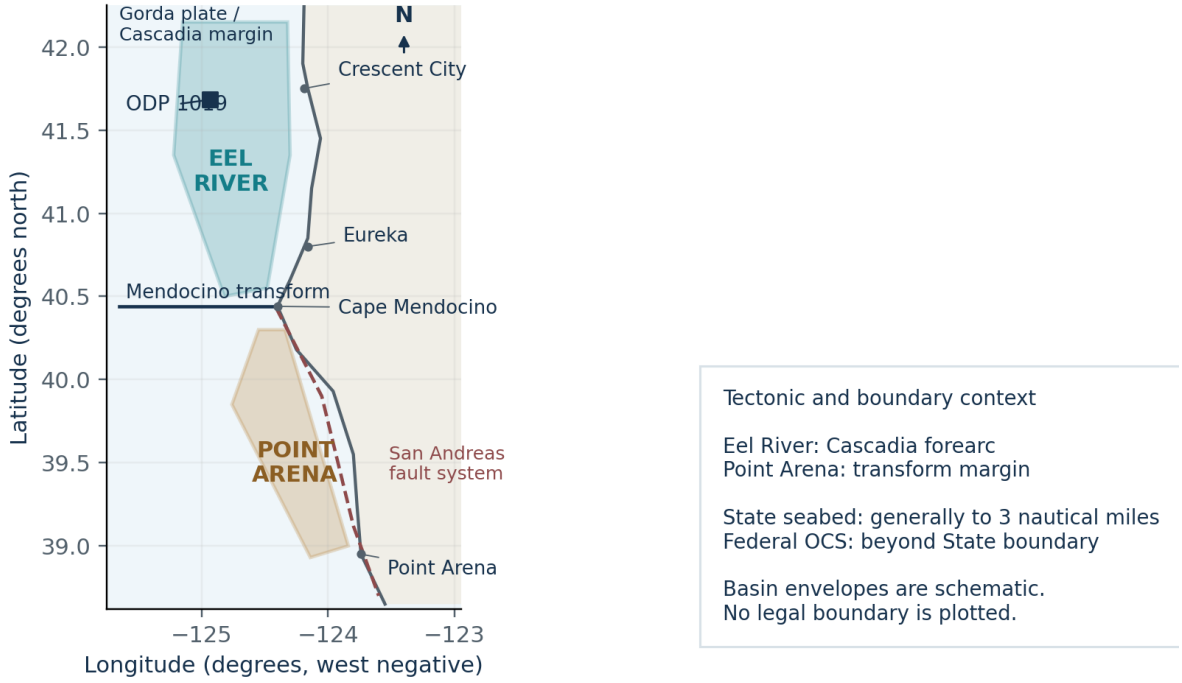


Figure 1. Generalized location and tectonic context, created from BOEM (2026), Figs. 18-19, and Furlong et al. (2024), Fig. 1 [1], [11]. Site 1019 position follows [7], p. 353. Outlines and fault traces are schematic.

Definition / source	Extent used	Uncertainty / implication
BOEM Eel River [1], 2026	Offshore forearc north of Cape Mendocino, with onshore continuation near Eureka.	Assessment outline differs from the exposed onshore basin and crosses planning-area boundaries.
CGS onshore Eel River [2], 2006	Humboldt structural depression, with the Little Salmon thrust on the northeast and Russ fault to the south.	Subsurface basement geometry is extrapolated, not demonstrated by a basement penetration.
BOEM Point Arena [1], 2026	Shelf/slope basin south of Cape Mendocino; small coastal extensions near Point Delgada and Point Arena.	Offshore assessment area is broader than the coastal stratigraphic sections in [4].
State / Federal [13]	State submerged lands generally extend to three nautical miles offshore. Federal OCS lies beyond the State boundary.	Jurisdiction does not coincide with a sedimentary-basin edge. The legal line is not drawn on Figure 1.

Table 1. Boundary definitions and the spatial support of the evidence [1], [2], [4], [13].

The northeast Eel River margin lies against accreted Franciscan rocks and local thrust structures. Point Arena lies within the San Andreas transform-margin setting. Regional crustal models require local seismic and well ties [2], [11].

3. Basin evolution

Basin evolution determines when organic matter was deposited, how deeply it was buried, and whether later deformation created or destroyed storage. The regional history is consequently part of the petroleum model rather than background description. Eel River records forearc deposition above accreted basement. The coastal Point Arena succession records an older sedimentary section and a distinct Oligocene-Miocene depositional history [3], [4].

Age / stage	Evidence and event	Petroleum significance
Pre-Neogene framework	Franciscan accretionary terranes underlie Eel River. Point Arena coastal outcrops include Paleogene sedimentary rocks [2]-[4].	Basement relief and older structures influence basin accommodation and later migration. The two areas cannot share one assumed basement history.
Oligocene-Miocene, Point Arena	Biostratigraphy distinguishes the Iversen, Skooner Gulch, Gallaway and Point Arena succession and a substantial older hiatus [4].	Separate unconformities from apparently continuous burial. Offshore source maturity cannot be inferred from coastal formation presence alone.
Early-middle Miocene, Eel River	Older upper Cenozoic sequence represents slope-basin and slope-blanket deposits [3].	Deep-water sandstone and enclosing mudstone provide candidate reservoir/seal pairs, with variable distribution.
Late Miocene-Pleistocene, Eel River	Younger marine sequence develops as forearc-basin fill, followed locally by shallowing deposits [2], [3].	Sedimentary loading supplies burial and overburden, while lateral facies changes create reservoir heterogeneity.
Late Neogene, Gorda Escarpment	Seismic ties and an unconformity support emergence near 3.7 Ma, greatest emergence near 2.7 Ma and return to sea level near 2.5 Ma [9], section 5.	Uplift, erosion and renewed subsidence can interrupt maturation and reorganize pathways. These ages apply to the studied ridge, not every Point Arena prospect.
Quaternary-present	Southern Cascadia anticlines, thrusts and seep pathways remain closely related [8].	Trap growth can coexist with leakage. Present closure does not prove that closure preceded the main charge.

Table 2. Age-event-significance synthesis. Ages reproduce each cited study’s temporal terminology; historical epoch labels are not used to redefine modern chronostratigraphic boundaries.

The most defensible interpretation is spatially variable burial and preservation. A section that remained in a depocentre could receive more thermal exposure than an equivalent unit later uplifted along the basin margin. The Gorda Escarpment record supplies a concrete example of this distinction. Its dated unconformity is evidence for local vertical motion, not a basin-wide charge date [9].

Useful authoritative frameworks are BOEM’s paired assessment maps [1] and CGS’s onshore depth-to-basement map [2], Fig. 8. CGS explicitly describes the latter as an extrapolation from a shallower structure map. It is suitable for regional architecture, with uncertainty increasing away from well control.

4. Stratigraphic framework

Table 3 is a simplified framework, not a claim that formations correlate continuously across the Mendocino transition. Petroleum roles are assigned to lithology and demonstrated occurrences. A formation can contain reservoir sandstone and sealing mudstone in different intervals.

Area / unit	Age, lithology and setting	Supported role / thickness context
Eel River: young cover, Carlotta and Scotia Bluffs	Pleistocene shallow-marine to nonmarine sand, gravel and associated fines [2].	Overburden and shallow aquifers. No deep petroleum reservoir is established simply by surface exposure.
Eel River: Rio Dell and Eel River formations	Primarily Pliocene marine sandstones, siltstone and mudstone; fan to shelf facies [2].	Demonstrated gas reservoir intervals onshore. Enclosing mudstone provides local seal candidates. Individual beds vary greatly in thickness.
Eel River: Pullen Formation	Upper Miocene and younger basal Wildcat marine siliceous/clastic facies [2], [3].	Candidate reservoir/seal succession. CGS reports gas shows, rather than established production, for Pullen sandstones.
Eel River: Bear River beds	Lower-middle Miocene deep-water sandstone enclosed in siliceous mudstone [2].	Candidate reservoir/seal pair. Distribution is restricted and affected by unconformities.
Eel River: Yager complex / older rocks	Paleogene deep-marine mudrocks and turbidites in [5]. The age terminology in [2] conflicts with this assignment.	Candidate gas-prone source and fractured older substrate. No single offshore source correlation is established here.
Point Arena: Point Arena Formation	Miocene siliceous marine succession, treated as Monterey-equivalent in the coastal study [4].	Possible source and fracture reservoir intervals. Actual oil shows support petroleum occurrence but not a quantified source kitchen [1].
Point Arena: Gallaway, Skooner Gulch, Iversen and German Rancho succession	Older marine clastic units, basalt and Paleogene strata separated by hiatuses [4].	Pre-Monterey sandstone targets are assessed offshore [1]. Basalt is a stratigraphic marker, not an assumed petroleum source.

Table 3. Simplified stratigraphic summary, ordered broadly from young to old within each area. [2]-[5] provide the geological basis; [1] defines the assessed offshore play names.

Thickness and correlation controls

CGS compiles a lower Wildcat marine succession about 1,890-2,500 m thick and an upper nonmarine succession about 800-1,000 m thick [2], p. 33. These are stratigraphic-package ranges in the onshore compilation, not net reservoir pay or a uniform offshore thickness. Coastal Point Arena formation thicknesses are not assigned to the entire offshore assessment area because a verified, continuous well-to-outcrop correlation is unavailable in the accessed evidence.

The Yager age discrepancy is retained explicitly. This report follows the dedicated Yager study’s Paleogene terminology [5] and records the Cretaceous designation in the older compilation [2] as an unresolved nomenclatural/age inconsistency. Resolving that issue requires original mapping and biostratigraphic ties, rather than averaging the two descriptions.

5. Source rocks, maturation and charge

Organic matter, maturity and correlation answer different questions. Organic carbon documents material available for hydrocarbon generation. Maturity assesses thermal alteration. A source-to-fluid correlation tests which interval actually charged an accumulation. The evidence assembled here supports hydrocarbon-generating processes, but leaves a specific offshore kitchen-to-trap pairing uncertain.

Candidate / evidence	Measurement or interpretation	What the evidence establishes
Yager structural complex [5]	Study abstract describes dominant Type III kerogen, relatively low organic carbon/yield and mature shales. Numerical Rock-Eval and reflectance tables were not accessible.	A plausible gas-prone source candidate. It does not establish that Yager charged a named offshore well. Qualitative supporting use only.
Shallow Neogene/Quaternary marine organic matter [7]	Hole 1019C usually has 0.5-1.5 wt% TOC, with an uppermost sample near 2.2 wt%. TOC/TN generally suggests marine organic matter.	Measured organic carbon at one shallow site. These data do not measure the richness or maturity of a deeply buried source kitchen.
Eel River gas occurrences [6]	USGS describes both thermogenic and microbial hydrocarbons in its regional gas-sampling study. No isotope values are transcribed without the original table.	Multiple origins are credible. A single genetic label for all field gas or all seeps would exceed the accessible evidence.
Point Arena siliceous units / shows [1], [4]	BOEM reports oil shows and interprets silica-diagenetic reflectors as indicators relevant to generation and fracture reservoirs.	Petroleum occurrence and an assessed source/reservoir concept. Reflector interpretation is not a direct TOC, kerogen or maturity measurement.

Table 4. Source-rock evidence and maturity assessment. TOC = total organic carbon; TOC/TN = organic carbon/total nitrogen. No basin-wide TOC, hydrogen index, Tmax or vitrinite-reflectance range is invented.

A defensible charge interpretation

Eel River offers a working gas-system interpretation with both shallow microbial production and deeper thermogenic contributions as possibilities. Burial of marine fines supplies a mechanism for source development, while older mudrocks provide an additional candidate. Present gas composition can also reflect mixing and alteration during migration. Establishing the dominant source at Tompkins Hill or a particular offshore trap would require formation-specific source extracts and paired gas/oil geochemistry [5], [6].

Point Arena supports a more conditional thermogenic oil/gas model. A siliceous source interval and its diagenetic history are consistent with the assessed plays, but a measured thermal-history model calibrated to a named well is not available in this evidence set. Consequently, neither a basin-wide oil window nor an absolute expulsion date is assigned [1].

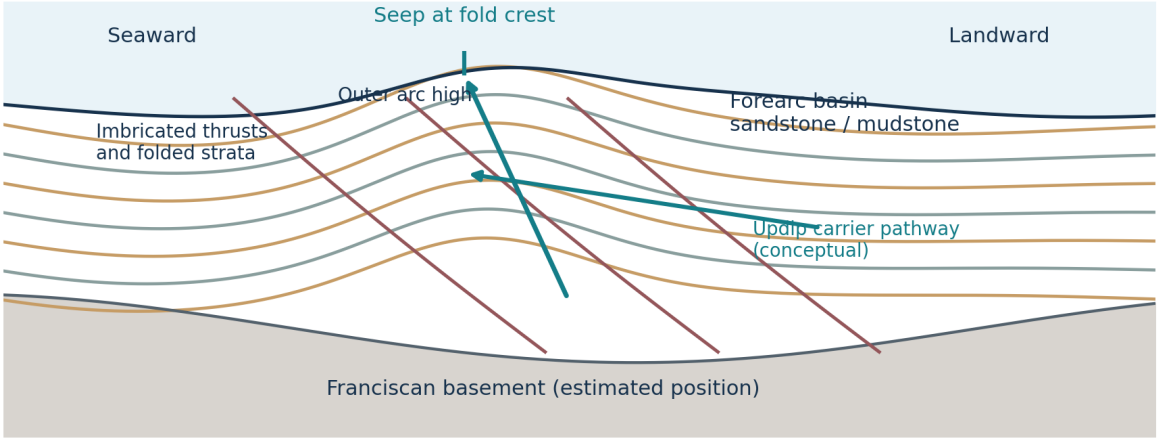
Source kitchen, expulsion and trap timing are the weakest connected part of the model. Additional maturation measurements would be most useful if obtained along a seismic section that ties a deeply buried source interval to a reservoir and an independently dated seal/trap geometry.

6. Reservoirs, seals and traps

Onshore production demonstrates that some Eel River sandstone-mudstone packages can retain gas. The offshore extension must additionally satisfy reservoir continuity, effective permeability and seal preservation. Fractures can provide permeability while also connecting a reservoir to leakage paths.

Target	Published properties / occurrence	Trap, seal and limitation
Rio Dell sandstone, onshore Eel River [2]	Historical compilation: porosity 22-28%, permeability 1-12 mD. Tompkins Hill production intervals about 640-1,770 m.	Anticlinal closure and flanking stratigraphic traps with enclosing marine fines. Values are compiled onshore data, not offshore basin averages.
Older / deeper Eel River sandstones [1], [2]	Fan, shelf and Paleogene sandstone plays. Gas shows occur in some older onshore units.	Lateral pinch-outs and structural closure are candidates. Cementation, fines and disconnected sand bodies could impair deliverability.
Point Arena sandstone and fractured siliceous plays [1]	BOEM assesses Neogene sandstone, Monterey fractured and pre-Monterey sandstone plays. No verified basin-wide porosity/permeability range is available.	Closure must coincide with charge and sealing facies. Fracture connectivity and fault sealing require local tests.

Table 5. Reservoir-trap comparison. mD = millidarcy. [2] compiles earlier field/formation reports; the underlying core analyses have not been remeasured.



Qualitative adaptation of Rudebusch et al. (2023), Fig. 6C. Geometry is not to scale. Arrows are inferred pathways.

Figure 2. Subsurface framework adapted qualitatively from the published southern Cascadia model in Rudebusch et al. (2023), Fig. 6C [8]. Original Fig. 6A-B shows the associated seismic framework. This redraw preserves the conceptual relationships, not seismic picks, scale or prospect geometry.

The published framework places a deformed outer arc high seaward of the forearc basin. Folds can focus updip movement, while thrust faults and near-vertical pathways convey fluids towards seep sites. This explains why an anticline can be both a migration focus and a leaking structure. It does not demonstrate a continuously sealed hydrocarbon column at every fold [8].

7. Migration, timing and preservation

The integrated source-to-trap sequence begins with organic-bearing sedimentation and burial. Thermogenic fluids, where generated, must leave a source interval and enter a connected carrier bed or fault. Buoyancy and pressure gradients can then move them towards structural highs. Accumulation requires an effective capillary seal and sufficient closure at the time of charge. The final step is preservation through continued faulting and uplift.

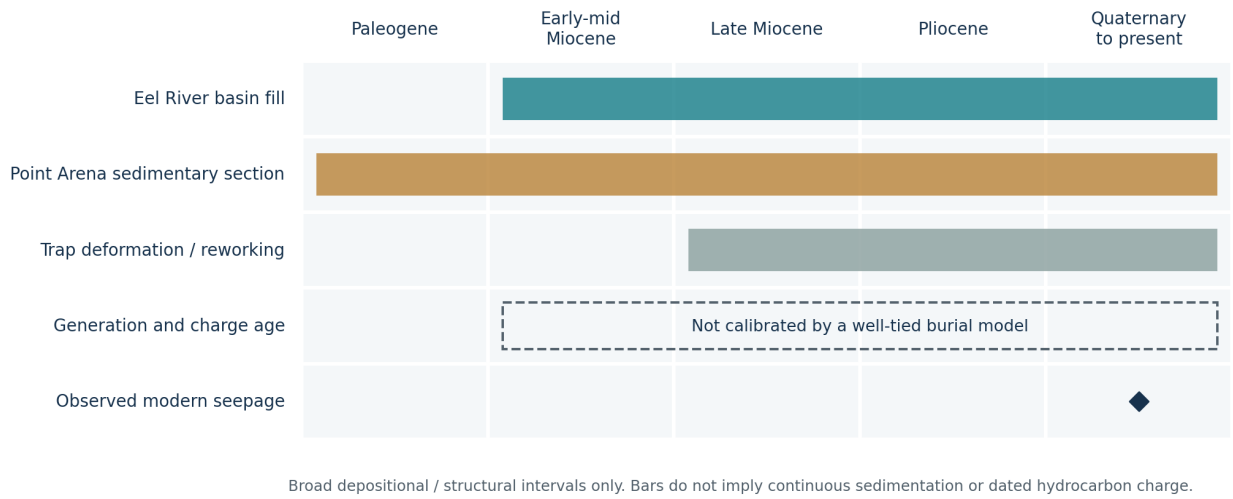


Figure 3. Relative petroleum-system events chart created from [1]-[5], [8] and [9]. Depositional and deformation bars are broad geological context. The dashed charge interval explicitly records missing age calibration; it is not a modeled generation history.

Eel River: demonstrated migration, uneven preservation

The local petroleum observations fit a compartmentalized gas system. Productive sandstone traps exist onshore, whereas offshore fluid escape is conspicuous along structures. Rudebusch et al. link seep locations to anticlines and faults using bathymetry and seismic interpretation. Those pathways are observationally constrained at studied locations, but their deeper connection to a specific thermogenic source remains interpretive [2], [8].

Microbial activity at a seep must be interpreted carefully. Orphan et al. document organisms and lipid signatures associated with anaerobic methane oxidation. That process consumes methane arriving at the seep; it does not prove that the gas originally formed by microbial methanogenesis. Gas-origin assignment therefore requires the gas geochemistry described by a source study rather than the presence of a microbial mat alone [6], [12].

Point Arena: charge and closure require a well tie

Oil shows establish that hydrocarbons occurred in the sampled exploration system. They do not reveal how much petroleum was retained or whether reservoir charge followed trap formation. Sandstone carriers and fractured siliceous intervals are plausible pathways within the assessed plays. Their migration geometry is a basin-model interpretation until a source interval, fault network and charged reservoir can be tied together by seismic and well evidence [1].

8. Representative evidence locations

Five evidence cases test different parts of the model. The selection includes a productive field, a small unsuccessful accumulation, an offshore well with limited hydrocarbon evidence, active seepage and a Point Arena exploration group. This avoids selecting only examples that confirm prospectivity.

Location / case	Observed evidence	Test of the model / limitation
Tompkins Hill field, onshore Eel River [2]	Rio Dell sandstone gas production and structural/stratigraphic trapping in the historical CGS compilation.	Demonstrates functioning reservoir, charge and retention locally. It does not establish the source interval or offshore reservoir quality.
Table Bluff field, onshore Eel River [2]	A Rio Dell anticlinal accumulation produced only a small volume before abandonment in the historical account.	Shows that structural closure alone does not guarantee a large retained volume or sustained deliverability. The record is historical, not a 2026 operating-status claim.
OCS-Petty P 0019 #1, offshore Eel River [1]	BOEM identifies gilsonite veins in a core from the bottom of the well as the only hydrocarbon indication among the four offshore exploratory wells it summarizes.	Petroleum occurrence is compatible with migration, but drilling did not demonstrate a commercial offshore accumulation. Possible failure causes remain nonunique.
Southern Eel River / Cascadia seep structures [8], [12]	Active methane discharge, structurally focused fluid pathways and methane-oxidizing microbial communities.	Confirms migration and leakage at sampled sites. Seep gas flux, recoverable volume and reservoir productivity are different quantities.
Point Arena exploratory well group [1]	BOEM reports oil shows in three offshore exploratory wells drilled during the 1960s and in two onshore wells.	Supports an active petroleum history. Shows alone do not establish flow capacity, connected pore volume or an economic discovery.

Table 6. Evidence tests. Well counts and the Petty well identifier follow BOEM (2026), pp. 69-71. No additional well names, test rates or reservoir measurements are inferred.

What the contrasting outcomes mean

The onshore/offshore difference cannot be reduced to “charged” versus “uncharged.” A dry or unsuccessful structural test may reflect poor reservoir quality, inadequate charge, leakage, adverse timing or a target that missed the relevant interval. The evidence does not uniquely separate those explanations. The appropriate exploration response is a prospect-specific fault/seal and source correlation study, not a blanket rejection of the basin.

Similarly, extensive seepage can indicate ongoing supply while also showing that some seals fail. A prospect close to a seep could benefit from a connected migration route or suffer from pressure loss and leakage. A 3-D structural model and fluid samples from the target reservoir would distinguish those possibilities.

ODP Site 1019 is treated separately as the thermal control on the next page. Its shallow scientific coring programme was not a deep petroleum discovery test [7].

9. Thermal regime

Hole 1019C provides direct offshore subsurface temperature evidence. Depths below are metres below seafloor (mbsf), rather than water depth or metres below sea level. The downhole values come from fitting the temperature-tool records after penetration disturbance [7], p. 370, Figs. 19-20.

Location / depth	Temperature or heat-flow evidence	Method and limitation
1019C: 36.8, 55.8, 74.8 mbsf	5.8, 6.7 and 8.1°C, respectively [7].	Three downhole Adara determinations. Fitted tool records, not deep equilibrium petroleum-well BHTs.
1019C: seafloor datum	Bottom water 3.7 ± 0.1°C; published gradient 57°C/km and heat flow 57 mW/m² [7].	Gradient uses bottom water plus three downhole points. Heat flow is derived using measured mean conductivity 0.989 W/(m·K).
Regional Cape Mendocino / Coast Ranges	Observed regional heat-flow context and transient thermal calculations [10].	Regional compilation/model, not a Point Arena well temperature or local geothermal reservoir test.

Table 7. Local measurements and regional thermal context [7], [10].

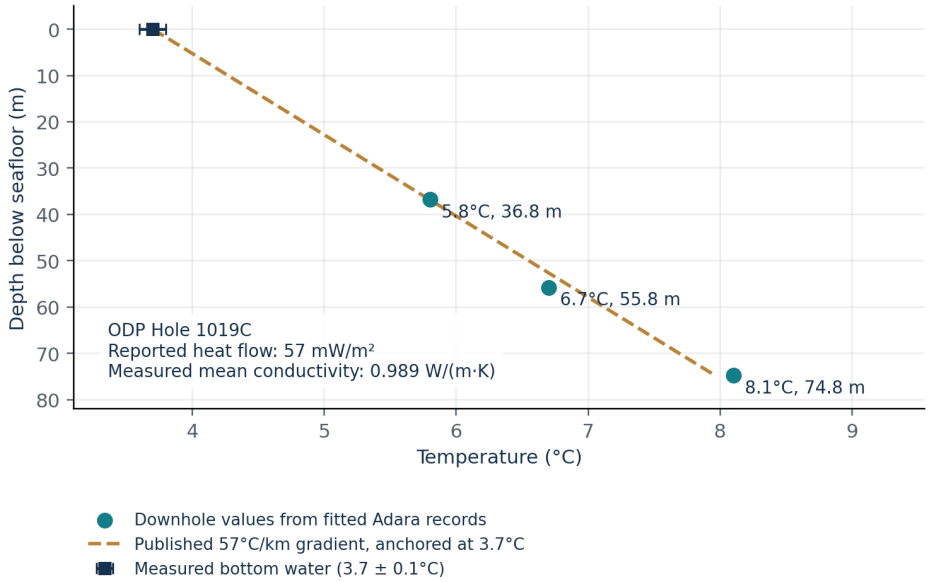


Figure 4. Temperature-depth data replotted from Shipboard Scientific Party (1997), p. 370; Figs. 19-20 [7]. The line depicts the published gradient over the observed shallow interval. It is not extrapolated to petroleum-reservoir depths.

Consistency calculation. Fourier conduction gives q = kG. Using the rounded published inputs, q = 0.989 × (57/1000) = 0.056373 W/m² = **56.4 mW/m²**. The source reports 57 mW/m². Both are retained rather than silently replacing the published estimate; the unrounded calculation was not reproduced.

A steep gradient in low-conductivity shallow sediment need not imply exceptional heat flux. Present shallow temperature also cannot substitute for cumulative thermal exposure during burial. Local conductivity, sedimentation, fluid flow and uplift must enter any deeper maturity model [7], [10].

10. Geothermal significance

The geothermal assessment is a screening interpretation from the evidence, not a resource estimate. A usable geothermal system requires recoverable heat and a sustainable means of moving fluid through it. The reviewed publications provide a measured shallow thermal anchor and regional tectonic context, but no demonstrated reservoir-scale thermal production test.

Concept	Evidence-supported assessment	Decisive missing information
Conventional hydrothermal	A productive hot-fluid reservoir is not demonstrated in the reviewed focus-area evidence.	Deep equilibrium temperature, reservoir pressure, fluid chemistry, connected permeability and sustainable flow.
Sedimentary-basin geothermal	Sandstone reservoirs make circulation a physical possibility. Their historical gas performance cannot determine thermal-water yield.	Well-tied temperature/conductivity profile, hydraulic tests and reinjection connectivity.
Coproduction from petroleum infrastructure	A conditional engineering concept for suitable wells. Offshore development concepts in BOEM do not establish an available geothermal installation.	Actual usable well condition, produced-water temperature/rate, lift and treatment requirements, access and heat demand.
Enhanced geothermal concepts	The regional tectonic setting motivates research questions but supplies no site-specific design basis.	Deep rock temperature, stress field, injectivity, connected stimulation volume and induced-seismicity assessment.
Commercial offshore project	No defensible commercial-resource declaration follows from the accessed evidence.	Offshore access, installation and connection costs, temperature/flow performance, environmental constraints and project-specific permissions.

Table 8. Preliminary geothermal screening. Physical and engineering conclusions are inferences from the data limitations, with local thermal evidence in [7], regional modeling in [10], and offshore infrastructure context in [1].

Thermal history and spatial variability

Guzofski and Furlong compare regional heat flow with models that combine transient crustal deformation and mantle upwelling. Their work shows why the passage of the Mendocino triple junction can change thermal conditions with location and time [10]. The modeled crustal system is much larger and deeper than a petroleum reservoir, so its temperatures cannot be inserted directly into a reservoir economic calculation.

More recent tomography emphasizes a three-dimensional plate-boundary architecture, including lithospheric heterogeneity south of the triple junction [11]. This strengthens the case for spatially variable thermal histories and weakens any assumption that one geothermal gradient represents both study basins.

The most useful next thermal test

A targeted programme would obtain a corrected, equilibrated temperature-depth log and conductivity profile in a well with a documented stratigraphic tie. Pressure-transient and fluid-chemistry measurements should be collected at the same interval. That combination would test whether sufficient heat, transmissivity and fluid supply coexist. Only then would a recovery or power estimate have a defensible physical basis.

11. Uncertainty, implications and conclusions

Uncertainty / severity	Consequence	Evidence that would reduce it
Source-to-fluid correlation / high	Petroleum occurrence does not identify the contributing source unit or relative microbial/thermogenic proportions.	Paired source-rock extracts and reservoir/seep gas or oil geochemistry with isotopes and biomarkers.
Generation versus trap timing / high	A present-day closure may have formed after the main charge or have leaked during later deformation.	Well-calibrated burial/maturity model tied to growth strata and fault-seal history.
Offshore reservoir continuity / high	Onshore sandstone properties and fractured-rock play concepts may not predict connected offshore flow.	Core, petrophysics, pressure communication and interval flow tests.
Deep thermal regime / high	One shallow offshore gradient cannot fix deeper temperatures or sustainable geothermal output.	Equilibrium temperature logs, conductivity and hydraulic tests at target depths.
Boundary / stratigraphic transfer / moderate	Assessment outlines, outcrop sections and onshore structural maps describe different spatial supports.	Consistent seismic/well framework, explicit boundary metadata and revised age correlations.

Table 9. Uncertainty register and proposed discriminating evidence. These are limitations of the present interpretation, not proof of low prospectivity.

Conclusions

1. The assigned focus area spans forearc and transform-margin settings. Eel River and Point Arena require distinct stratigraphic and thermal histories, with their offshore comparison grounded in BOEM assessment definitions [1], [11].

2. Eel River has demonstrated onshore sandstone gas accumulations and offshore fluid migration. Those observations support an operating petroleum system locally while leaving offshore source correlation and retained accumulation size unresolved [2], [6], [8].

3. Point Arena oil shows support petroleum occurrence and assessed sandstone/fractured siliceous plays. They do not establish a uniformly mature source kitchen, productive fracture network or commercial discovery [1], [4].

4. Folding and faulting can create trapping geometry and concentrate migration while also allowing leakage. Charge and preservation must therefore be evaluated against the timing of deformation [8], [9].

5. Site 1019 provides a genuine shallow thermal measurement, with a reported 57°C/km gradient and derived 57 mW/m² heat flow. Deep maturation and geothermal conclusions require additional well-specific evidence. A commercial geothermal resource cannot be declared from the present evidence [7], [10].

Scientific scope: literature-based screening. No new reserve calculation, calibrated burial model, geothermal power estimate or claim of student-performed source inspection is implied.

References

Numbered citation style. Technical source count: 12, including 8 peer-reviewed journal articles, 3 agency technical reports and 1 scientific drilling report. The additional jurisdiction webpage [13] is not included in that count. All links accessed October 20, 2026.

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[12] V. J. Orphan, K.-U. Hinrichs, W. Ussler III, C. K. Paull, L. T. Taylor, S. P. Sylva, J. M. Hayes, and E. F. DeLong, “Comparative analysis of methane-oxidizing archaea and sulfate-reducing bacteria in anoxic marine sediments,” *Applied and Environmental Microbiology*, vol. 67, no. 4, pp. 1922-1934, 2001. doi:10.1128/AEM.67.4.1922-1934.2001. https://pmc.ncbi.nlm.nih.gov/articles/PMC92814/

[13] California State Lands Commission, “About the California State Lands Commission,” Land Management Division description, accessed October 20, 2026. https://www.slc.ca.gov/about/

Appendix A1. Prompt / research log

AI assistance was used for literature discovery, comparison, drafting, figure/table preparation and technical critique. The entries summarize the questions actually addressed during this assisted workflow; they are not a claimed verbatim transcript. Scientific support comes from the listed publications and datasets, not the AI tool.

Assignment prompt	Research question addressed	Recorded output
1. Define the study area	Compare Eel River, Point Arena, onshore Humboldt, assessment province and State/Federal definitions.	Section 2 and Figure 1.
2. Build the source library	Find and audit technical literature with direct agency/publisher identifiers.	References and Appendix A5.
3. Geologic history	Link deposition, subsidence, uplift and triple-junction migration to petroleum processes.	Section 3, Table 2.
4. Stratigraphy	Keep the two basin successions distinct and qualify petroleum roles.	Section 4, Table 3.
5. Source and maturity	Separate organic richness, maturity, fluid origin and demonstrated correlation.	Section 5, Table 4.
6. Reservoir / seal / trap	Keep compiled onshore properties separate from offshore play concepts.	Section 6, Table 5 and Figure 2.
7. Migration and timing	Build a connected source-to-trap argument and identify missing charge dates.	Section 7 and Figure 3.
8. Thermal regime	Inspect original Site 1019 temperatures, depth datum, fitting method and conductivity.	Section 9, Table 7 and Figure 4.
9. Geothermal significance	Test whether heat, transmissivity, fluids and infrastructure support a resource claim.	Section 10, Table 8.
10. Skeptical review	Challenge basin-wide extrapolation, seep interpretation, trap timing and unverified source data.	Section 11 and Appendix A4.
11. Reference audit	Check authors, titles, report identifiers, DOI/official links, and claim-specific locators.	References and Appendix A2-A5.

Evidence and review conventions

Observation: a reported measurement, sample, well result or mapped feature. **Interpretation:** a publication’s geological explanation. **Inference:** a conclusion drawn in this report from the available evidence. Figures 1-3 include schematic interpretation; Figure 4 replots reported temperatures.

Aly Sadek confirmed personal verification of all twelve audited claims and their cited source sections on **October 20, 2026**. The student verification column records this attestation. Assistant inspection and access limitations are recorded separately; abstract-only access and unavailable tables describe assistant access.

AI-use disclosure: research, synthesis, drafting and document production were AI assisted. No fieldwork, laboratory analysis, instructor approval or student source review is fabricated.

Appendix A2. Claim-source verification

Locators use printed publication page numbers, not PDF viewer indices. Student verification: Aly Sadek attested all twelve claims on October 20, 2026. Assistant checks are recorded separately.

Load-bearing claim	Exact source / locator	Evidence and assistant check	Student verified?
1. The offshore assessment pair spans different geological provinces.	[1] sections 7.4.2-7.4.3, pp. 69-71; Figs. 18-19.	Agency interpretation Original report inspected	Yes
2. Onshore Eel River margins include the Little Salmon thrust and Russ fault.	[2] section 3.2.4.1, p. 32.	Mapped framework Original text inspected	Yes
3. Point Arena coastal stratigraphy differs from the Eel River forearc succession.	[3] author abstract; [4] pp. 419-422, text-Figs. 1-2; [2] pp. 32-33.	Stratigraphic interpretation [2] original; [3] abstract; [4] indexed original text	Yes
4. CGS compiles Rio Dell porosity 22-28% and permeability 1-12 mD.	[2] p. 33, Rio Dell paragraph. Underlying DOG/Stanley data are a secondary compilation.	Historical property compilation Original compilation inspected	Yes
5. The four offshore Eel River wells summarized by BOEM provide only limited hydrocarbon evidence.	[1] p. 69; OCS-Petty P 0019 #1 gilsonite-vein paragraph.	Well-result summary Original report inspected	Yes
6. Point Arena has oil shows in three offshore exploratory wells.	[1] p. 70, paragraph after area description.	Well-result summary Original report inspected	Yes

Audit rule

A bibliographic match establishes that a publication exists. It does not establish that a particular claim is supported. Numerical claims above use inspected original report text or tables. Abstract-only sources are retained for limited qualitative context and are explicitly labeled.

No values from paywalled or inaccessible TOC, Rock-Eval, vitrinite-reflectance or gas-isotope tables have been reconstructed. The absence of such values in this report records an evidence limit rather than a zero value or proof that no source potential exists.

Appendix A3. Claim-source verification (continued)

Locators use printed publication page numbers, not PDF viewer indices. Student verification: Aly Sadek attested all twelve claims on October 20, 2026. Assistant checks are recorded separately.

Load-bearing claim	Exact source / locator	Evidence and assistant check	Student verified?
7. Southern Cascadia faults and folds focus fluid flow and seepage.	[8] sections 3.3-3.4 and 4; Figs. 4-6.	Observation plus interpretation Original full text / figure caption inspected	Yes
8. The Eel River gas study allows both thermogenic and microbial origins.	[6] official USGS author abstract. Detailed isotopic table not accessed.	Published synthesis Official abstract only; numerical isotope claims excluded	Yes
9. 1019C downhole values are 5.8, 6.7 and 8.1°C at 36.8, 55.8 and 74.8 mbsf.	[7] p. 370, Heat Flow; Figs. 19-20, p. 376.	Measured records with fitted temperatures Original chapter inspected	Yes
10. 1019C has a reported 57°C/km gradient and measurement-based 57 mW/m² heat flow.	[7] p. 370; mean measured conductivity 0.989 W/(m·K).	Derived from measurements Original chapter and dimensional calculation inspected	Yes
11. Most 1019C TOC values are 0.5-1.5 wt%, with an uppermost value near 2.2 wt%.	[7] p. 369 and Table 14, p. 373; 2.18 wt% at 0.29 mbsf in printed table.	Measured sediment geochemistry Original chapter / printed table inspected	Yes
12. Gorda Escarpment emergence near 3.7-2.5 Ma is a local history, not a basin-wide charge date.	[9] section 5, especially Figs. 10-11 and unconformity discussion.	Seismic/sample interpretation Original full text inspected	Yes

Audit rule

A bibliographic match establishes that a publication exists. It does not establish that a particular claim is supported. Numerical claims above use inspected original report text or tables. Abstract-only sources are retained for limited qualitative context and are explicitly labeled.

No values from paywalled or inaccessible TOC, Rock-Eval, vitrinite-reflectance or gas-isotope tables have been reconstructed. The absence of such values in this report records an evidence limit rather than a zero value or proof that no source potential exists.

Appendix A4. Corrections and reflection

Statement challenged	Correction made after source checking
“Site 1019 has four downhole Adara measurements.”	The final chapter lists three downhole determinations and a bottom-water datum, giving four points for the gradient [7], p. 370. The report and plot use that distinction.
“A high offshore gradient establishes geothermal power potential.”	57°C/km is a shallow local gradient in low-conductivity sediment. The reported heat flux is 57 mW/m², and no deep flow test is supplied [7]. Resource language was narrowed to screening.
“A microbial mat means the seep methane was generated microbially.”	The cited microbiological paper measures methane oxidation [12]. Gas generation and gas consumption are different processes. Origin remains constrained by separate geochemistry [6].
“One Yager age can be copied without checking.”	[2] calls the Yager Formation Cretaceous; [5] explicitly treats a Paleogene Yager complex. The discrepancy is recorded, and no unsupported numerical burial history is built from it.
“The heat-flow calculation exactly reproduces the published result.”	The rounded inputs give 56.4 mW/m², while the chapter reports 57 mW/m². The report preserves the source value and discloses the calculation difference.

These are corrections or qualifications made during the assisted synthesis and audit. They are not presented as independently performed student actions.

Reflection

The assisted workflow was most useful for separating questions that initially appeared to be one basin study. It organized boundary definitions, linked petroleum-system elements to different kinds of evidence, and made it easier to compare productive fields with unsuccessful offshore tests. It also located a basin-specific thermal chapter with actual downhole values, which made a measured temperature-depth figure possible. The evidence tables exposed several places where a fluent explanation could have become stronger than its support. In particular, methane-consuming microbes cannot identify the original generation mechanism of the gas, and a shallow sediment gradient cannot establish deep geothermal output. Checking the original drilling chapter also corrected the measurement count and clarified the depth datum. Human geological judgment remains necessary to evaluate source correlation, seismic continuity, fault behavior and the timing of charge relative to trap growth. Those decisions require examining the original figures and deciding whether a regional interpretation is transferable to a particular prospect. The report therefore keeps interpretations and proposed tests distinct from measurements, retains negative exploration evidence, and avoids calculating a resource from missing reservoir data. The audit records assistant inspection separately from personal student review, so the workflow remains transparent about who performed each check.

Reflection length: 199 words. The assignment requests 150-250 words.

Appendix A5. Source library and access audit

Full citations and DOI/report numbers are in References. This inventory supplies source type, topic, why it matters and access limitations. “Direct” identifies evidence within the focus area; “regional” indicates contextual transfer.

Source / type	Topic and role	Scope / inspection limit
[1] Agency technical report	Assessment boundaries, offshore wells and plays	Direct, both basins. Original report, sections 7.4.2-7.4.3 and Figs. 18-19
[2] Agency technical report	Onshore structure, stratigraphy, reservoir compilation	Direct, onshore Eel River. Original report, section 3.2.4.1 and Figs. 8-9
[3] Peer-reviewed article	Two Neogene sequences and regional basin architecture	Direct, Eel River. Author abstract and metadata on USGS record. Detailed paywalled figures not used
[4] Peer-reviewed article	Point Arena formation names, biostratigraphy and missing section	Direct coastal outcrops, offshore extrapolation limited. Publisher-indexed original text and metadata. Full page images unavailable
[5] Peer-reviewed article	Yager age terminology, gas-prone organic matter and maturity	Direct outcrop study, buried source correlation unproven. Publisher metadata and abstract indexed by OSTI. Numerical tables not used
[6] Agency technical report	Gas occurrences, microbial and thermogenic contributions	Direct local gas sampling. Official USGS abstract/metadata. Original PDF could not be opened; no isotope numbers used
[7] Scientific drilling technical report	Measured temperatures, conductivity, organic carbon and depth datum	Direct, one shallow offshore site. Original chapter, Heat Flow and Organic Geochemistry sections; Figs. 19-20 and Table 14
[8] Peer-reviewed article	Seep structure, imbricated thrusts, faults and fluid pathways	Direct southern Cascadia, broader than Eel River. Original full text, sections 3.3-3.4 and 4; Figs. 4-6
[9] Peer-reviewed article	Seismic stratigraphy, unconformity and uplift/subsidence	Direct Gorda Escarpment / Vizcaino Block, regional to Point Arena. Original full text, sections 4-5; Figs. 7-11
[10] Peer-reviewed article	Regional observed heat-flow comparison and transient thermal modeling	Regional onshore Coast Ranges, not a Point Arena reservoir test. Original full text, paragraphs 6-12; Figs. 1-4
[11] Peer-reviewed article	Modern tomography and 3-D transition from subduction to translation	Regional tectonic framework. Original full text, sections 1.1-1.2 and 3
[12] Peer-reviewed article	Methane consumption, seep geochemistry and biological alteration	Direct Eel River seep samples with other-basin comparison. Original full text, methods and results; Tables 1-2
[13] Agency jurisdiction webpage	State submerged-land jurisdiction	California-wide boundary context. Official agency text

References [1]-[12] satisfy the substantial technical-source inventory. [13] supports jurisdiction only. None is an AI-generated scientific reference. Original figures/maps to inspect: [1] Figs. 18-19; [2] Figs. 8-9; [7] Figs. 19-20; [8] Fig. 6; [9] Figs. 7-11.